10B 1 HALOGENOALKANES AND HYDROLYSIS REACTIONS

SPECIFICATION REFERENCE
10.4 10.6 10.7

LEARNING OBJECTIVES

- Understand the nomenclature of halogenoalkanes and be able to draw their structural, displayed and skeletal formulae.
- Understand the distinction between primary, secondary and tertiary halogenoalkanes.
- Know the definition of the term 'nucleophile'.

WHAT ARE HALOGENOALKANES?

The halogenoalkanes are a homologous series of compounds with the general formula $C_nH_{2n+1}X$. Think of them as the result of replacing a hydrogen atom in a hydrocarbon by a halogen atom. X represents a halogen atom – usually bromine or chlorine, but it could also be fluorine or iodine. You have already met halogenoalkanes earlier in this book – chloromethane is a halogenoalkane and is the product of the reaction between methane and chlorine.

The symbol R is often used in organic chemistry to represent any alkyl group (such as methyl or ethyl). The formula of a halogenoalkane could be simplified to RX.

The number of halogen atoms in a halogenoalkane molecule can be more than one, so the general formula is different for these compounds (for example, $C_nH_{2n}X_2$ and $C_nH_{2n-1}X_3$ are other possible general formulae).



▲ **fig A** The most problematic halogenoalkanes are those containing both chlorine and fluorine. They are known as chlorofluorocarbons (CFCs for short) and were used in aerosol cans for decades. We have known for a long time that they damage the ozone layer in the upper atmosphere and, through international agreement, they are mostly no longer used.

NAMING HALOGENOALKANES

We have already met the names of some halogenoalkanes, but **table A** shows some examples to remind you.

EXAM HINT

Practise drawing skeletal formulae for halogenoalkanes. This will help you identify the different isomers for a given molecular formula.

STRUCTURE	NAME
$CH_2Cl–CHCl–CH_3$	1,2-dichloropropane
$CH_2Br–CH_2–CH_2Cl$	1-bromo-3-chloropropane
CCl_4	tetrachloromethane
$CH_3–CHF–CH_2–CH_3$	2-fluorobutane

table A Examples of how to name halogenoalkanes.

If there are two or more different halogens, then the prefixes used to represent the halogens appear in alphabetical order. As with other homologous series, the longest carbon chain is the basis of the name, and when the prefix numbers are added together the total should be as small as possible.

CLASSIFYING HALOGENOALKANES

You have met the terms 'primary', 'secondary' and 'tertiary' in the topic referring to carbocations (see **Topic 5A.4**).

Halogenoalkanes are classified in a similar way, depending on the number of alkyl groups joined to the C atom bonded to the halogen atom. **Table B** shows some examples.

STRUCTURE	ABBREVIATED FORMULA	NUMBER OF ALKYL GROUPS	CLASSIFICATION
$CH_3–CH_2–CH_2F$	RX	1	primary
$CH_3–CHBr–CH_3$	R_2CHX	2	secondary
$(CH_3)_2CCl–CH_2–CH_3$	R_3CX	3	tertiary

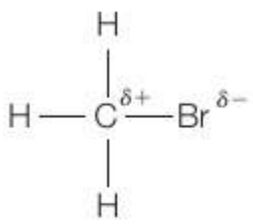
table B Examples of how to classify halogenoalkanes.

WHAT MAKES HALOGENOALKANES REACTIVE?

Hydrocarbons contain only hydrogen and carbon atoms, which have similar electronegativities, so their bonds are almost non-polar.

Halogenoalkanes contain a halogen atom with an electronegativity higher than that of carbon, so the C—X bond is polar. This bond polarity can be indicated using the partial charges $\delta+$ and $\delta-$.

Down Group 7 of the Periodic Table, the electronegativities of the halogens decrease from fluorine to iodine, so the polarity of the C—X bond also decreases.



The carbon atom joined to the halogen is always the slightly positive ($\delta+$) or electron-deficient part of the molecule, and this is what makes halogenoalkanes react as they do.

These carbon atoms attract other species called **nucleophiles**. 'Nucleo' indicates positive and 'phile' indicates liking, so nucleophiles are species that are attracted to slightly positive or electron-deficient parts of a molecule. Nucleophiles are either negative ions or molecules with a slightly negative atom, but they always use a lone pair of electrons when attacking another species.

We will look at the mechanisms of halogenoalkane reactions in **Topic 10B.3**, but first let us take a look at a reaction that can be used to compare the reactivities of different halogenoalkanes.

HYDROLYSIS REACTIONS

When a halogenoalkane is added to water, a reaction begins, but it may take some time to complete. A water molecule contains polar bonds, and the $\delta-$ oxygen atom in water is attracted to the $\delta+$ carbon atom in the halogenoalkane. The reaction that occurs can be represented by the equation below, in which R represents any alkyl group:

$$RX + H_2O \rightarrow ROH + HX \quad \text{or} \quad RX + H_2O \rightarrow ROH + H^+ + X^-$$

The product ROH is an alcohol and, as both organic substances are usually colourless liquids, no colour change can be seen. You can see from this simple equation that the C—X bond breaks, so the RX molecule breaks into two parts (R and X), although the R group then combines with the OH group of water.

This type of reaction is known as a **hydrolysis reaction**. 'Hydro' refers to water and 'lysis' refers to splitting, so it means splitting with water.

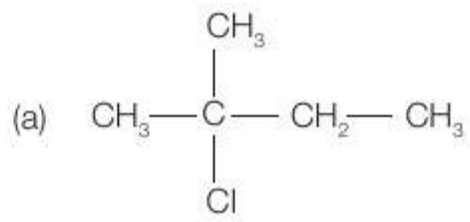
LEARNING TIP

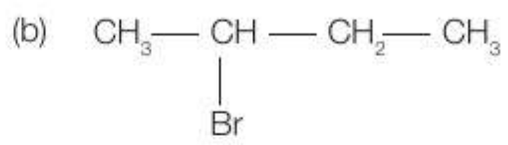
In an organic formula, the symbol R can be used to represent any alkyl group, such as methyl or ethyl.

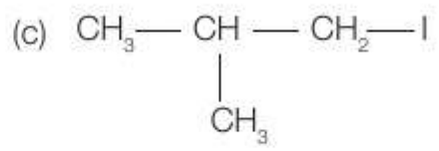
This should not be confused with the same symbol R used in a completely different context for the gas constant in the equation $pV = nRT$.

CHECKPOINT

1. Write the IUPAC name for each of these halogenoalkanes and classify them as primary, secondary or tertiary.







2. Why are nucleophiles more strongly attracted to fluoroalkanes than to chloroalkanes?

SUBJECT VOCABULARY

nucleophile a species that donates a lone pair of electrons to form a covalent bond with an electron-deficient atom

hydrolysis reaction a reaction in which water or hydroxide ions replace an atom in a molecule with an –OH group

10B 2 COMPARING THE RATES OF HYDROLYSIS REACTIONS

SPECIFICATION REFERENCE
10.10 10.12
10.13 CP5

LEARNING OBJECTIVES

- Understand that experimental observations and data can be used to compare the relative rates of hydrolysis of primary, secondary and tertiary structural isomers of a halogenoalkane, and primary chloro-, bromo- and iodo-alkanes, using aqueous silver nitrate in ethanol.
- Know the trend in reactivity of primary, secondary and tertiary halogenoalkanes.
- Understand, in terms of bond enthalpy, the trend in reactivity of chloro-, bromo- and iodo-alkanes.

PRACTICAL ASPECTS

If instead of adding water in a hydrolysis reaction, silver nitrate solution is added, then the progress of the reaction can be followed. You may remember that silver nitrate can be used in a test for halide ions because the silver ions in silver nitrate react with the halide ions formed in the hydrolysis to give a precipitate:

$$Ag^+ + X^- \rightarrow AgX$$

This means that we can tell how quickly the hydrolysis reaction occurs by observing how quickly the precipitate of AgX forms.

Without going into full practical details, a comparison of the rates of these reactions involves:

- using ethanol as a solvent for the mixture (halogenoalkanes and aqueous silver nitrate do not mix, but form separate layers)
- controlling variables such as temperature and the concentration and quantity of halogenoalkanes
- timing the appearance of the precipitate (although this is difficult to do accurately because a precipitate may first appear faint, but become thicker with time).

In an experiment, three tubes containing silver nitrate dissolved in ethanol are left at the same temperature for several minutes. Each tube contains a different halogenoalkane. **Fig A** shows the tubes after several minutes.



▲ **fig A** Three tubes containing silver nitrate dissolved in ethanol, with a different halogenoalkane added to each one.

Fig A shows that there is no precipitate in tube (a), a faint cream-coloured precipitate in tube (b) and a thicker pale yellow precipitate in tube (c). These observations are caused by the reaction of a chloroalkane in (a), a bromoalkane in (b) and an iodoalkane in (c).

Two types of comparison can be made. You can:

- compare halogenoalkanes with the same structure but containing different halogens
- compare halogenoalkanes containing the same halogen but with different structures.

Fluoroalkanes are comparatively very unreactive, so reactions involving them are often omitted.

INTERPRETING THE RESULTS FOR DIFFERENT HALOGENS

Table A shows the trend when the halogen is different.

SAME STRUCTURE BUT DIFFERENT HALOGEN	RESULT
1-iodobutane	fastest
1-bromobutane	
1-chlorobutane	slowest

table A Examples comparing halogenoalkanes with the same structure but with different halogens.

You might suppose that the halogenoalkane with the most polar bond would be the fastest to be hydrolysed (in this case, 1-chlorobutane) because the $\delta+$ charge on the carbon atom is greatest, so the attacking nucleophile should be attracted more strongly.

This is true, but there is another, more important factor to consider. Bond breaking requires energy, and weaker bonds break more easily than stronger bonds. **Table B** shows (mean) bond enthalpies. You can see that the C—I bond is the weakest and the C—Cl bond is the strongest.

BOND	BOND ENTHALPY / kJ mol^{-1}
C–Cl	+346
C–Br	+290
C–I	+228

table B Mean bond enthalpies.

Under the same conditions, the C—I bond breaks most easily, forming I^- ions, and so a precipitate of AgI forms more quickly.

The C—F bond is much stronger (+467 kJ mol^{-1}) than any of the others, which explains why fluoroalkanes are not often used in these hydrolysis experiments.

INTERPRETING THE RESULTS FOR DIFFERENT STRUCTURES

Table C shows the trend when the structure is different.

SAME HALOGEN BUT DIFFERENT STRUCTURE	RESULT
2-bromo-2-methylpropane (tertiary)	fastest
2-bromobutane (secondary)	
1-bromobutane (primary)	slowest

table C Examples comparing halogenoalkanes with the same halogen but with different structures.

To explain why tertiary halogenoalkanes are more rapidly hydrolysed than secondary and primary compounds requires a detailed understanding of two different reaction mechanisms, and is beyond the scope of this book, but will be dealt with in **Topic 15A (Book 2: IAL)**.

LEARNING TIP

Molecules with partial charges ($\delta+$ and $\delta-$) are not ions. They are molecules with no overall charge but with electron-rich and electron-deficient parts.

CHECKPOINT

1. 2-bromopropane and 2-iodopropane are put into separate test tubes and warmed with water. Explain which one is hydrolysed more quickly.
2. Why are fluoroalkanes not readily hydrolysed compared with chloroalkanes?

10B 3 HALOGENOALKANE REACTIONS AND MECHANISMS

SPECIFICATION REFERENCE
10.8 10.9

LEARNING OBJECTIVES

- Understand the reactions of halogenoalkanes with aqueous potassium hydroxide, aqueous silver nitrate in ethanol, potassium cyanide, ammonia and ethanolic potassium hydroxide.
- Understand the mechanisms of the nucleophilic substitution reactions between primary halogenoalkanes and aqueous potassium hydroxide and ammonia.

SUBSTITUTION REACTIONS

The hydrolysis reactions in **Topics 10B.1** and **10B.2** involved replacing a halogen atom (X) with a hydroxyl group (OH). These are substitution reactions, and here is a summary of the four reactions you need to know.

Reaction 1: RX $\xrightarrow{H_2O/\text{warm}}$ ROH

Reaction 2: RX $\xrightarrow{KOH/\text{heat under reflux}}$ ROH

Reaction 3: RX $\xrightarrow{KCN/\text{heat under reflux}}$ RCN

Reaction 4: RX $\xrightarrow{NH_3/\text{heat in sealed tube}}$ RNH_2

REACTION 1

This is the hydrolysis reaction you met in **Topic 10B.1**.

REACTION 2

Heating a halogenoalkane with aqueous potassium hydroxide under reflux is one way of making alcohols. The attacking nucleophile is the OH^- ion. An example of an equation for this reaction is the conversion of 1-chloropropane into propan-1-ol:

$$CH_3CH_2CH_2Cl + KOH \rightarrow CH_3CH_2CH_2OH + KCl$$

or

$$CH_3CH_2CH_2Cl + OH^- \rightarrow CH_3CH_2CH_2OH + Cl^-$$

The advantage of an ionic equation is that the nucleophile is clearly shown.

REACTION 3

Heating a halogenoalkane with potassium cyanide dissolved in ethanol under reflux is one way of making **nitriles**. The attacking nucleophile is the CN^- ion. An example of an equation for this reaction is the conversion of bromoethane into propanenitrile:

$$CH_3CH_2Br + KCN \rightarrow CH_3CH_2CN + KBr$$

or

$$CH_3CH_2Br + CN^- \rightarrow CH_3CH_2CN + Br^-$$

Note that the organic product contains one more carbon atom than the starting material. This reaction is a useful way of increasing the length of the carbon chain. It is also an important way of synthesising more complex compounds.

EXAM HINT

If you were to carry out this reaction in the lab then there would need to be a very careful risk assessment. KCN is highly toxic!

REACTION 4

Heating a halogenoalkane with ammonia solution under pressure in a sealed tube is one way of making **primary amines**. The sealed tube is needed because ammonia is a gas and would otherwise escape from the apparatus before it could react. The attacking nucleophile is the NH_3 molecule. An example of an equation for this reaction is the conversion of 1-iodobutane into butylamine:

$$CH_3CH_2CH_2CH_2I + NH_3 \rightarrow CH_3CH_2CH_2CH_2NH_2 + HI$$

This equation looks similar to those for Reactions 1–3, but it isn't quite the full story. Like NH_3, the organic product is a base, so it would react with the inorganic product, the acid HI, to form a salt. So a better equation is:

$$CH_3CH_2CH_2CH_2I + NH_3 \rightarrow CH_3CH_2CH_2CH_2NH_3^+ + I^-$$

However, this is only the first step because the product is a salt, not a primary amine. To produce a high yield of the amine, the ammonia is used in excess, and some of this excess ammonia reacts in a second step to produce the amine:

$$CH_3CH_2CH_2CH_2NH_3^+I^- + NH_3 \rightarrow CH_3CH_2CH_2CH_2NH_2 + NH_4^+I^-$$

The final products are butylamine and ammonium iodide. These two steps are often combined as:

$$CH_3CH_2CH_2CH_2I + 2NH_3 \rightarrow CH_3CH_2CH_2CH_2NH_2 + NH_4^+I^-$$

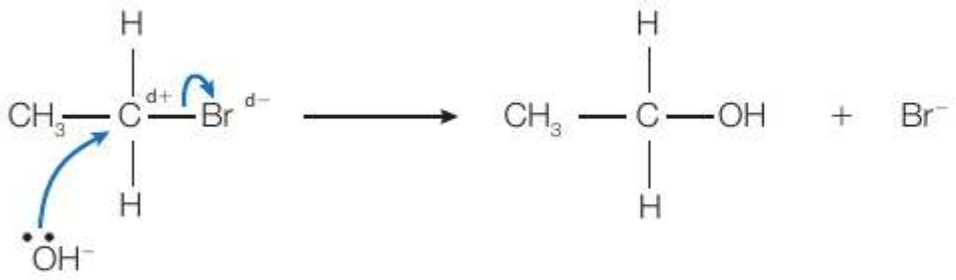
NUCLEOPHILIC SUBSTITUTION MECHANISMS

You have already learned about mechanisms for free radical substitution and electrophilic addition; here is a different mechanism to learn about.

In each reaction the attacking species is a nucleophile, so the reaction type is described as **nucleophilic substitution**. You only need to know about mechanisms for primary halogenoalkanes undergoing Reactions 2 and 4.

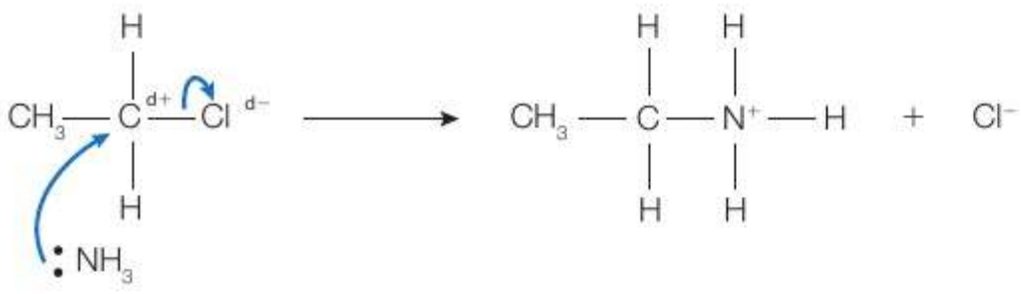
MECHANISM OF REACTION 2

One example is the reaction between bromoethane and aqueous potassium hydroxide. The reaction starts with the donation of a lone pair of electrons from the oxygen of a hydroxide ion to the electron-deficient carbon atom and the formation of a C—O bond. At the same time, the electrons in the C—Br bond move to the Br atom, resulting in the breaking of the C—Br bond. This type of bond breaking is known as heterolytic fission. 'Hetero' indicates different, and you already know that 'lysis' indicates breaking.

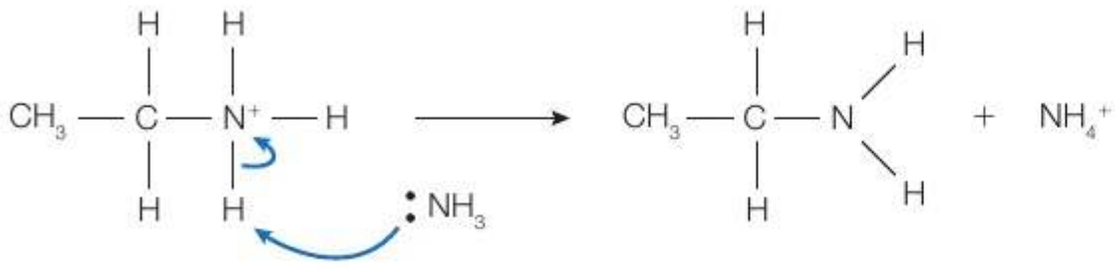


MECHANISM OF REACTION 4

One example is the reaction between chloroethane and ammonia. The first step of the reaction involves the donation of a lone pair of electrons from the nitrogen of an ammonia molecule to the electron-deficient carbon atom and the formation of a C—N bond. At the same time, the electrons in the C—Cl bond move to the Cl atom, resulting in the breaking of the C—Cl bond.



The second step of the reaction involves another ammonia molecule acting as a base and removing a hydrogen ion from the ion formed in the first step.



ELIMINATION REACTIONS

Reaction 2 above is a nucleophilic substitution reaction, but using a different solvent (ethanol instead of water) causes a different reaction to occur. When a halogenoalkane is heated with **ethanolic** potassium hydroxide, the OH^- ion acts as a base and not as a nucleophile. You know that a base reacts with a hydrogen ion (H^+), but in the case of a halogenoalkane, the hydrogen that reacts with the OH^- ion is the one attached to a carbon atom next to the C in the C—Br bond.

For example, the equation for the reaction between 2-bromopropane and ethanolic potassium hydroxide is:

$$CH_3—CHBr—CH_3 + KOH \rightarrow CH_2=CH—CH_3 + H_2O + KBr$$

The organic product is propene (an alkene), and water and potassium bromide are the other products. You can see why the reaction is referred to as elimination. H and Br are removed from the halogenoalkane but they are not replaced by any other atoms.

You do not need to know the mechanism for this **elimination reaction**.

EXAM HINT

The fact that the same reagents can give different products depending on which solvent you use highlights the need for you to make sure that you know both reactants *and* conditions for an organic synthesis.

LEARNING TIP

Both the hydroxide ion and the ammonia molecule can act as nucleophiles and as bases.

CHECKPOINT

1. Chloromethane is heated with:

(a) ammonia

(b) potassium cyanide dissolved in ethanol.

Give the name of the organic product formed in each case.

2. 2-chlorobutane is heated with:

(a) aqueous potassium hydroxide

(b) ethanolic potassium hydroxide.

State the mechanism of each reaction and explain why there are two different organic products in (b).

SUBJECT VOCABULARY

nitrile organic compound containing the C–CN group

primary amine compound containing the $C–NH_2$ group

nucleophilic substitution a reaction in which an attacking nucleophile replaces an existing atom or group in a molecule

ethanolic a solution in which ethanol is the solvent

elimination reaction a reaction in which a molecule loses atoms attached to adjacent carbon atoms, forming a C=C double bond